<h1 id="Titanic-Exploratory-Data-Analysis">Titanic Exploratory Data Analysis</h1><h2 id="Purpose">Purpose</h2> I will explore the Titanic passenger dataset to understand survival patterns, missing data, and relationships between passenger details and survival.

Import the libraries

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
```

Load the data

In [3]:
```
df = pd.read_csv("train.csv")

print("Rows and columns:", df.shape)
df.head()
```

```
Rows and columns: (891, 12)
```

Out[3]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>PassengerId</th> <th>Survived</th> <th>Pclass</th> <th>Name</th> <th>Sex</th> <th>Age</th> <th>SibSp</th> <th>Parch</th> <th>Ticket</th> <th>Fare</th> <th>Cabin</th> <th>Embarked</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Check the columns and data types

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Count missing values and duplicate rows

In [6]:
```
df.isnull().sum()
```

Out[6]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

Look at number summaries

In [7]:
```
df.describe()
```

Out[7]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>PassengerId</th> <th>Survived</th> <th>Pclass</th> <th>Age</th> <th>SibSp</th> <th>Parch</th> <th>Fare</th> </thead> <tbody> <th>count</th> <th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50%</th> <th>75%</th> <th>max</th> </tbody> <tbody></tbody>

891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Count how many passengers survived

In [9]:
```
df["Survived"].value_counts().sort_index()
```

Out[9]:
```
Survived
0    549
1    342
Name: count, dtype: int64
```

Compare survival by sex and passenger class

In [10]:
```
print("Survival rate by sex (%):")
print((df.groupby("Sex")["Survived"].mean() * 100).round(1))

print("\nSurvival rate by passenger class (%):")
print((df.groupby("Pclass")["Survived"].mean() * 100).round(1))
```

```
Survival rate by sex (%):
Sex
female    74.2
male      18.9
Name: Survived, dtype: float64

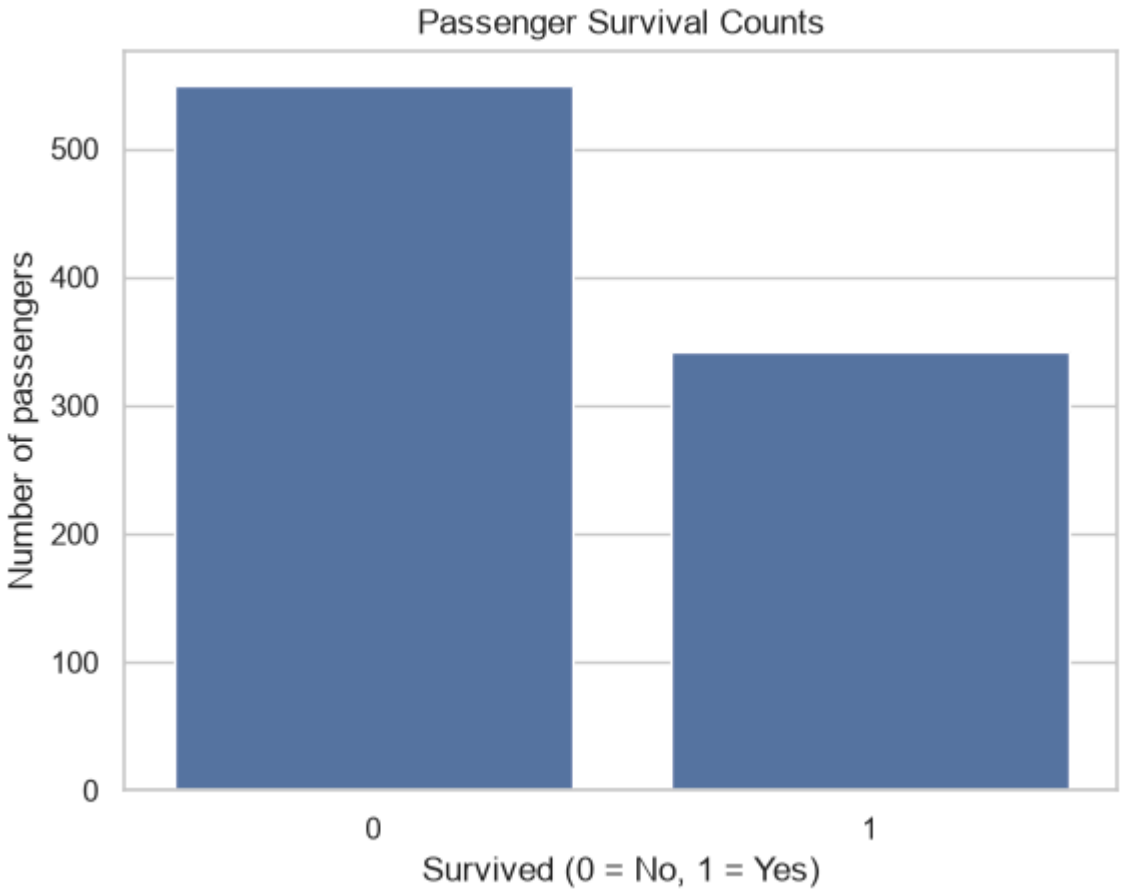
Survival rate by passenger class (%):
Pclass
1    63.0
2    47.3
3    24.2
Name: Survived, dtype: float64
```

Survival counts

In [11]:
```
import matplotlib.pyplot as plt
import seaborn as sns

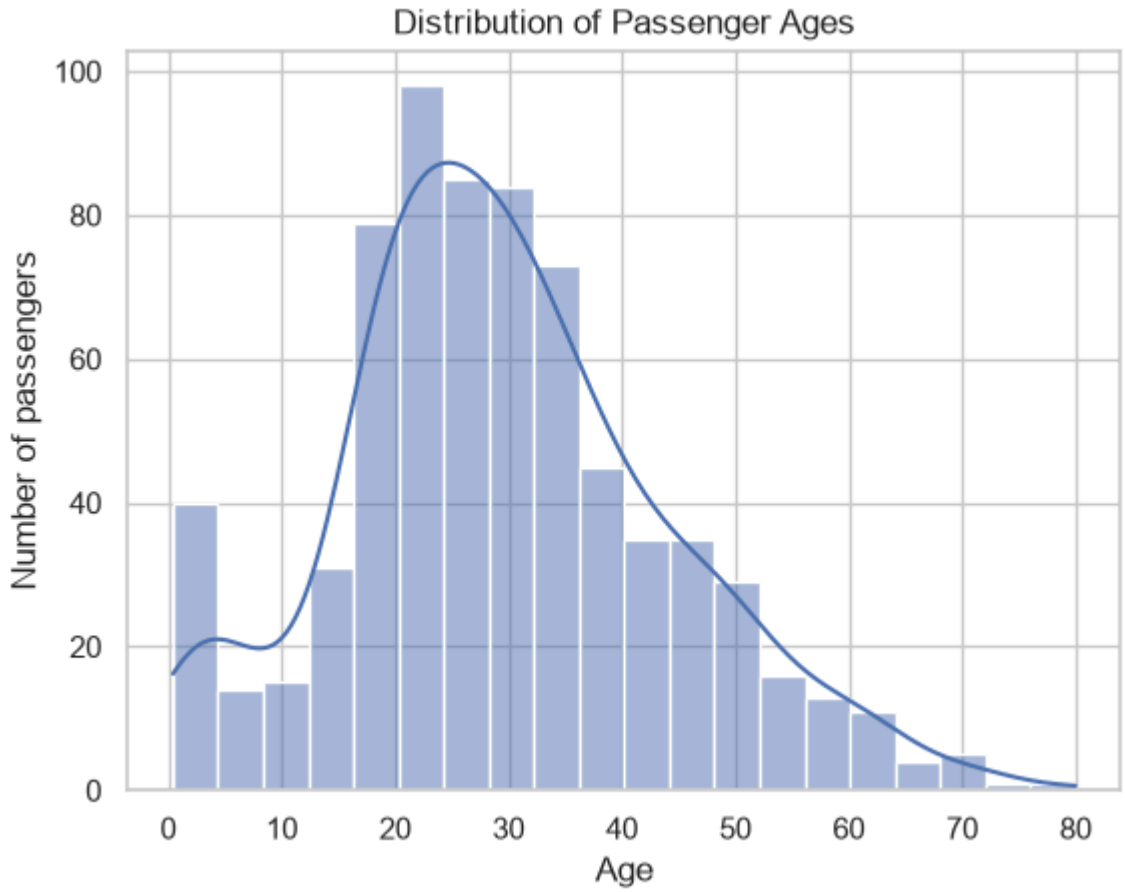
sns.countplot(data=df, x="Survived")
plt.title("Passenger Survival Counts")
```

```
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of passengers")
plt.show()
```



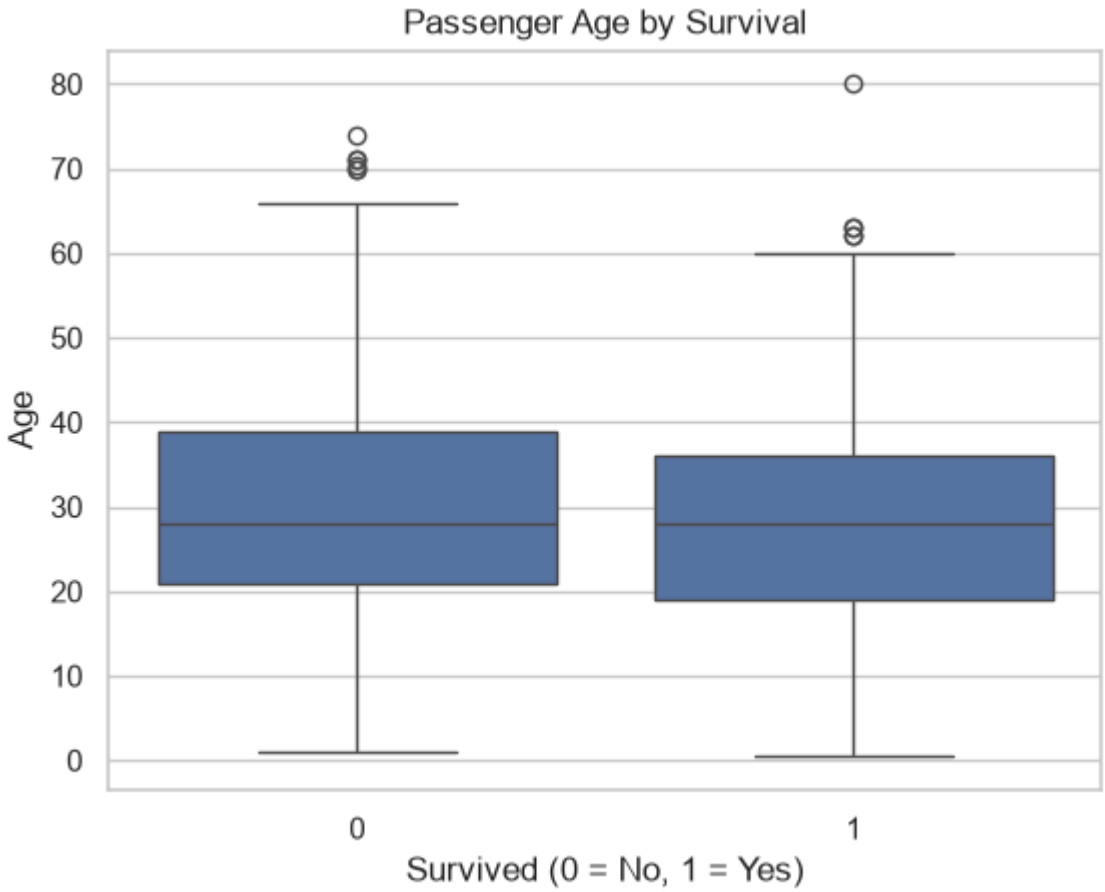
Age distribution

In [12]:
```
sns.histplot(data=df, x="Age", bins=20, kde=True)
plt.title("Distribution of Passenger Ages")
plt.xlabel("Age")
plt.ylabel("Number of passengers")
plt.show()
```



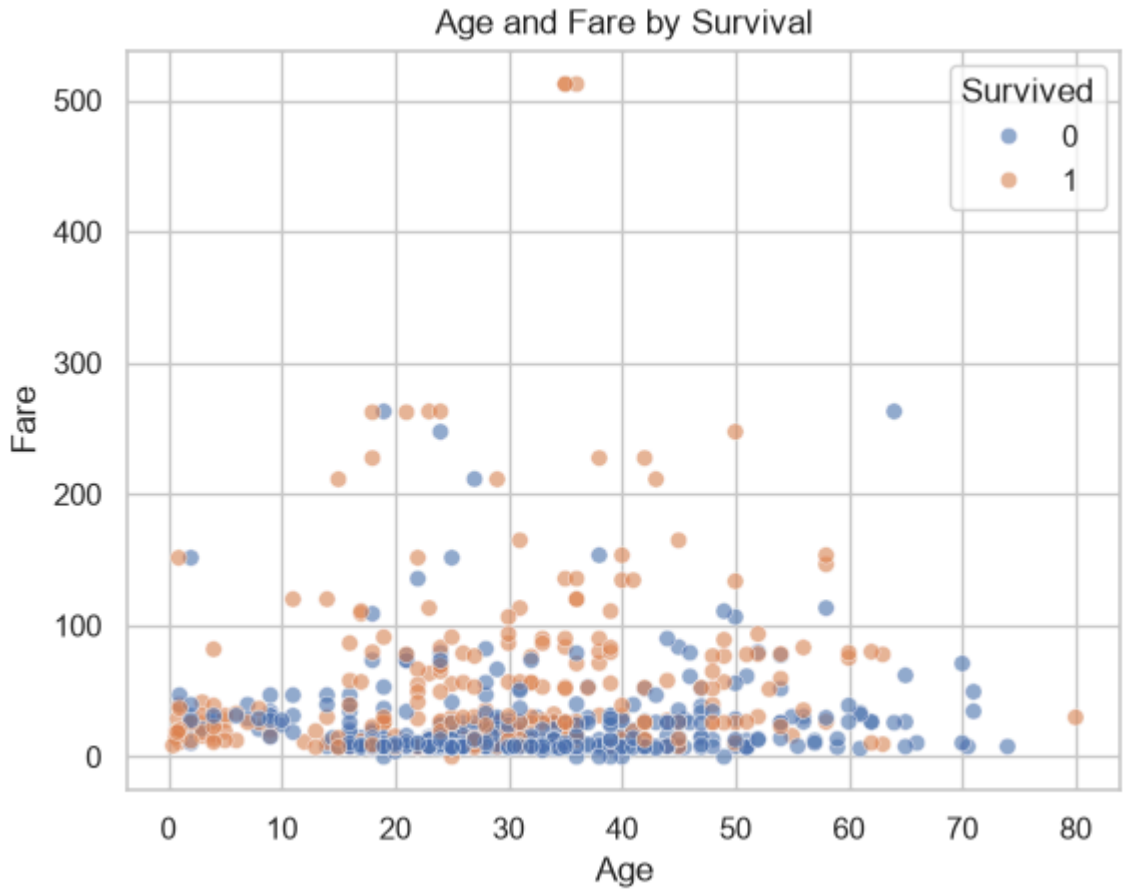
Age by survival

In [13]:
```
sns.boxplot(data=df, x="Survived", y="Age")
plt.title("Passenger Age by Survival")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Age")
plt.show()
```



Fare compared with age

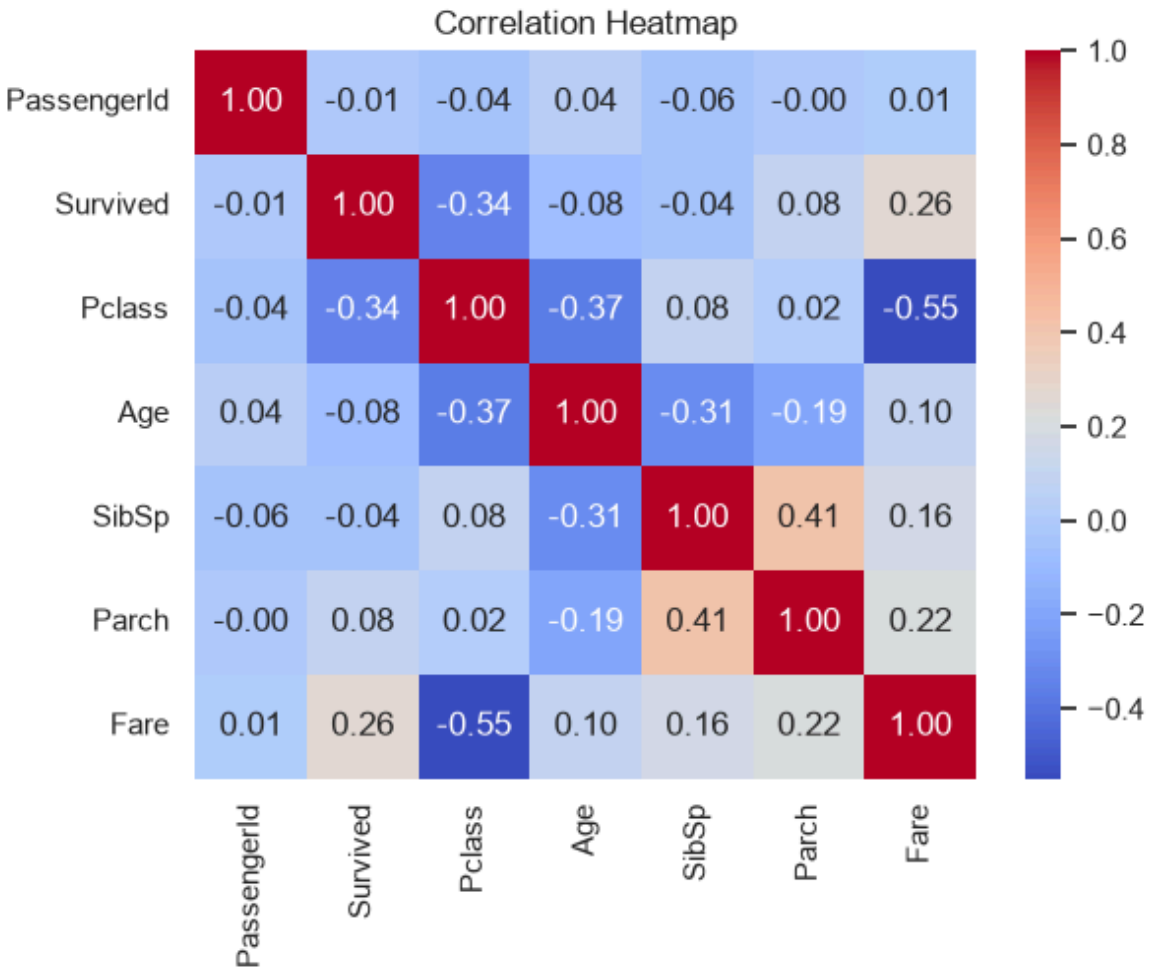
In [14]:
```
sns.scatterplot(data=df, x="Age", y="Fare", hue="Survived", alpha=0.6)
plt.title("Age and Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```



Heatmap of numeric-column relationships

In [17]:
```
numeric_data = df.select_dtypes(include="number")

sns.heatmap(numeric_data.corr(), annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```



<h2 id="Findings">Findings</h2>

- The dataset contains 891 passengers and 12 columns.
- The columns with missing information are Age, Cabin, and Embarked.
- The number of passengers who survived was ___; the number who did not survive was ___.
- The survival rate was higher for ___.
- The age chart shows ___.
- One other pattern I noticed was ___.

<h2 id="Conclusion">Conclusion</h2> This exploratory analysis found patterns in passenger age, fare, passenger class, sex, and survival. These are observations from this dataset and do not prove cause and effect.